Welcome to Code Crunch!

C4 | CRUNCH LABS: HACKATHON

THU 2-3PM Weekly | Spring 2025

Attendance

Earn a Certificate of Completion by attending at least 70% of the workshops

linktr.ee/CODE.CRUNCH

CODE CRUNCH

Unit #0

C4 | CRUNCH LABS: HACKATHON Overview

Welcome to Code Crunch!

- Code Crunch is your go-to platform for collaboration, networking, and hands-on projects. We welcome freshmen, graduates, and alumni from any university, regardless of major or experience level.

- Join us virtually via Zoom or in person at Florida International University. Together, we'll break down barriers in technology, learn through immersive workshops, and build innovative solutions.

Our Mission & Vision

Mission:

Our mission is to equip students with the technical skills and professional connections needed to excel in coding interviews and beyond. We provide a platform where students can collaborate, learn, and grow professionally while emphasizing the importance of diversity and inclusion in the tech industry

Vision:

We strive to bridge the gap between academia and the tech industry by offering hands-on workshops, technical interview preparation, hackathons, and networking opportunities with experts from leading companies. Our vision is to empower students from all backgrounds to achieve their full potential, foster an inclusive tech community, and build a strong, supportive alumni network. By covering a wide range of technical areas, we aim to prepare students for success in a variety of tech careers.

What We Offer

- Hands-On Workshops: Topics include Python, JavaScript, React, and more.
- LeetCode Prep: Weekly sessions to master technical interviews.
- Hackathons: Collaborate on innovative projects in a competitive setting.
- Panel Discussions: Industry leaders from tech giants and startups share insights.
- Global Access: Code Crunch Global launches in October for worldwide participation!

Meet the E-Board

- **President:** Brian Bazurto
- **Vice President:** Amilcar Pena
- **Treasurer:** Bryan Garcia
- **Marketing/Logistics:** Henry Cruz
- **Logistics:** Jason Liang

Tech Leads:

- Alvise Carpin
- Arnold Miranda
- Daniel Hernandez
- Gabriel Nogueras
- Justin Joseph
- Mariana Phelan
- Sara Salvatierra
- Thierry Laguerre

Why Join Code Crunch?

- Collaborate with peers and mentors.
- Develop technical and leadership skills.
- Network with professionals from top companies.
- Build impactful projects and gain hands-on experience.

A welcoming and inclusive space for underrepresented groups in tech.

Spring 2025 Weekly Schedule

Funded by SGA | Open to All Students

CODE CRUNCH WEEKLY SCHEDULE - SPRING 25

Cost: No-cost

January 6, 2025 - April 4, 2025

Your CS Club for Collaboration, Networking, and Hands-On Projects!

Open to all students, regardless of major—no experience needed! ***Freshmen, graduates, and alumni from any university are welcome***. Let's innovate together!

Crunch AI Data Science Python	**CodeCrunch LABS Portfolio** Java & React	**Wire Crunch: Embedded Systems** C++	**Crunch Convos** Python	**Cybersecurity Crunch** VM/LINUX
MON 1 PM - 2 PM WED 1 PM - 2 PM	MON 2 PM - 3 PM	TUE 1 PM - 2 PM	WED 2 PM - 3 PM	THU 1 PM - 2 PM
CodeCrunch LABS Hackathon Java & React	**CodeCrunch LABS: Web Dev** Javascript	**Crunch The Code** Python/ Leetcode	**Swe Convos** Networking	
THU 2 PM - 3 PM	THU 3 PM - 4 PM	FRI 1 PM - 2 PM	FRI 2 PM - 3 PM	

Attend 7 of 10 Weekly Sessions in the Same Workshop to Earn a Certificate of Completion!

RSVP on LU.MA/CODECRUNCH

In person or Stream on Zoom

GO.FIU.EDU/CODECRUNCH

LINKTR.EE/CODE.CRUNCH

Become a member

Be Part of the Code Crunch Community

1. RSVP for sessions at go.fiu.edu/codecrunch.
2. Follow us on LinkedIn, Instagram, and Discord for updates.
3. Become a member using the QR code on this slide.
4. Attend workshops, participate in events, and join our leadership team.

Together, we innovate and grow.

Let’s Innovate Together!

Thank you for joining us on this exciting journey! Code Crunch is redefining tech education and fostering inclusivity in technology. Let’s collaborate, innovate, and create a brighter future together.

RSVP for sessions today. Follow us on social media. Spread the word about Code Crunch!

#CodeCrunch #CodeCrunchFIU #DiversityInTech #InclusiveTech #LeadershipOpportunities #FIUTechCommunity #WomenInSTEM #LGBTQInTech #LatinxInTech #BlackInTech #AAPIsInTech #TechLeadership #JoinOurTeam #FIUInnovation #TechForEveryone #EmpowerTech #FutureLeaders #HackerCulture #CodeYourFuture #TechDiversity #FIUPanthers #STEMOpportunities #BuildYourFuture #LeadWithCode #PanthersInTech

C3 | CRUNCH LABS: PORTFOLIO OVERVIEW

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Java Fundamentals and Object-Oriented Programming Overview

Java Fundamentals & OOP Overview

Java Basics

- **Variables**: `int age = 20;` | `String name = "Arnold";`
- **Functions**: `public int add(int a, int b) { return a + b; }`
- **Loops**: `for`, `while`, `do-while`
- **`return` vs `print`**:
  `return` → sends value to caller
  `print` → displays output

Java Libraries

- Key packages: `java.lang`, `java.util`, `java.io`

OOP Principles

1. **Encapsulation** – Protect data using `private` fields + getters/setters
2. **Inheritance** – Reuse code (`Dog extends Animal`)
3. **Polymorphism** – One method, many forms
4. **Abstraction** – Hide complexity using interfaces/abstract classes

Agile Methodologies

Definition

Agile is a flexible, iterative approach to project management and software development, focused on delivering small, incremental changes quickly.

Key Components

1. **Sprints** – Short development cycles (1-4 weeks).
2. **Scrum Meetings** – Daily check-ins for progress updates.
3. **Backlog** – A prioritized list of tasks and features.
4. **User Stories** – Simple descriptions of features from the user's perspective.
5. **Retrospectives** – Meetings at the end of sprints to improve processes.

Importance

- **Adaptability**: Quickly responds to changes in requirements.
- **Collaboration**: Encourages teamwork and continuous feedback.
- **Efficiency**: Delivers working software faster.

Introduction to App Development

JavaScript & React Native for Frontend Development

- **JavaScript**: Core programming language for building interactive web applications.
- **React Native**: A framework for building mobile apps using JavaScript, enabling cross-platform development for iOS and Android with a shared codebase

Use Cases of Expo Go

1. **Rapid Prototyping** – Quickly build and test app features.
2. **Collaborative Development** – Share QR codes with teammates or clients to review progress.
3. **Simplified Testing** – Instantly test apps on multiple devices without platform-specific builds.

Why Use Expo Go?

- **No Native Builds Needed**: You can develop and test your app without having to build an APK or IPA.
- **Cross-Platform Development**: Works for both iOS and Android devices with a single codebase.
- **Live Reloading**: Automatically reloads the app when code changes are saved.
- **Access to Expo SDK**: Provides pre-built libraries for common tasks like camera access, location services, push notifications, and more.

Overview of Expo Go

Expo Go is a mobile app that allows developers to preview and test **React Native** apps directly on their physical devices without needing to build or install the app manually. It simplifies the development process by offering a fast and convenient way to see changes in real-time.

Creating a Log-in UI

React Native Components:

- Use `TextInput` for email/password fields and `Button` for login/signup actions.

Install Firebase

Use Firebase's `signInWithEmailAndPassword` method

Firebase Benefits for Authentication

- **Secure** and scalable authentication system.
- Supports **email/password**, **Google**, **Facebook**, and other providers.
- Real-time database integration for user-specific data.

Database and Management

Database Management in React Native

- **Options**:
  - **SQLite** – Local relational database, great for offline storage.
  - **Firebase Realtime Database** – Cloud-hosted, provides real-time sync.
  - **Realm** – Fast, object-based local database for large datasets.
- **Common Use Cases**:
  - Storing user preferences/settings.
  - Caching data for offline functionality.
  - Managing app-specific data (e.g., notes, to-do lists).

Benefits of Using UseContext & AsyncStorage

- **UseContext**:
  - Centralizes app-wide state (e.g., auth, theme).
  - Makes state sharing easier across components.
- **AsyncStorage**:
  - Provides persistent data storage even when the app is closed.
  - Ideal for small amounts of data like tokens, user preferences, etc.

Set Up an API

What is Strapi?

- Strapi is a "headless" Content Management System (CMS) that makes it easy to build APIs
- It provides a user-friendly admin panel to manage content
- Perfect for developers who need a quick way to create and manage APIs

Slide 2 - Why Use Strapi?

- Ready-to-use admin interface
- Customizable API endpoints
- Built-in user authentication
- Database integrations out of the box
- RESTful API and GraphQL support

Creating an API with Strapi

1. Access the admin panel (typically http://localhost:1337/admin)
2. Navigate to "Content-Types Builder"
3. Create your data structure (collections, fields)
4. Set permissions in "Settings > Users & Permissions"

App Display and UI

Key FlatList Properties

- `horizontal`: Enables horizontal scrolling
- `pagingEnabled`: Snaps to full width items
- `showsHorizontalScrollIndicator`: Hides/shows scroll bar
- `snapToInterval`: Controls sliding snap points

ScrollView Properties

- `contentContainerStyle`: Styles the content container
- `showsVerticalScrollIndicator`: Controls scroll bar visibility
- `bounces`: Controls bounce effect (iOS)
- `overScrollMode`: Controls over-scroll behavior (Android)

Navigate with React

Best Practices

- Always include loading states during API calls
- Implement proper error handling
- Use environment variables for API URLs
- Consider implementing request caching
- Add proper validation before sending data
- Include user feedback for successful/failed operations

Optional Video Player Integration and Export

APK Location & Testing

- Debug APK location: `android/app/build/outputs/apk/debug/app-debug.apk`
- Release APK location: `android/app/build/outputs/apk/release/app-release.apk`
- Testing steps:
  1. Install on Android device
  2. Test all functionality
  3. Verify permissions
  4. Check app performance

Q&A Session

Attendance

Join us for the upcoming weekly sessions this Spring 2025!

Monday: 1PM & 2PM
Tuesdays: 1PM
Wednesdays: 2PM
Thursdays: 1PM , 2PM & 3PM
Friday: 2PM & 3PM

RSVP LU.MA/CODECRUNCH

Your feedback helps us improve. Share your thoughts!
Go to our website: go.fiu.edu/codecrunch